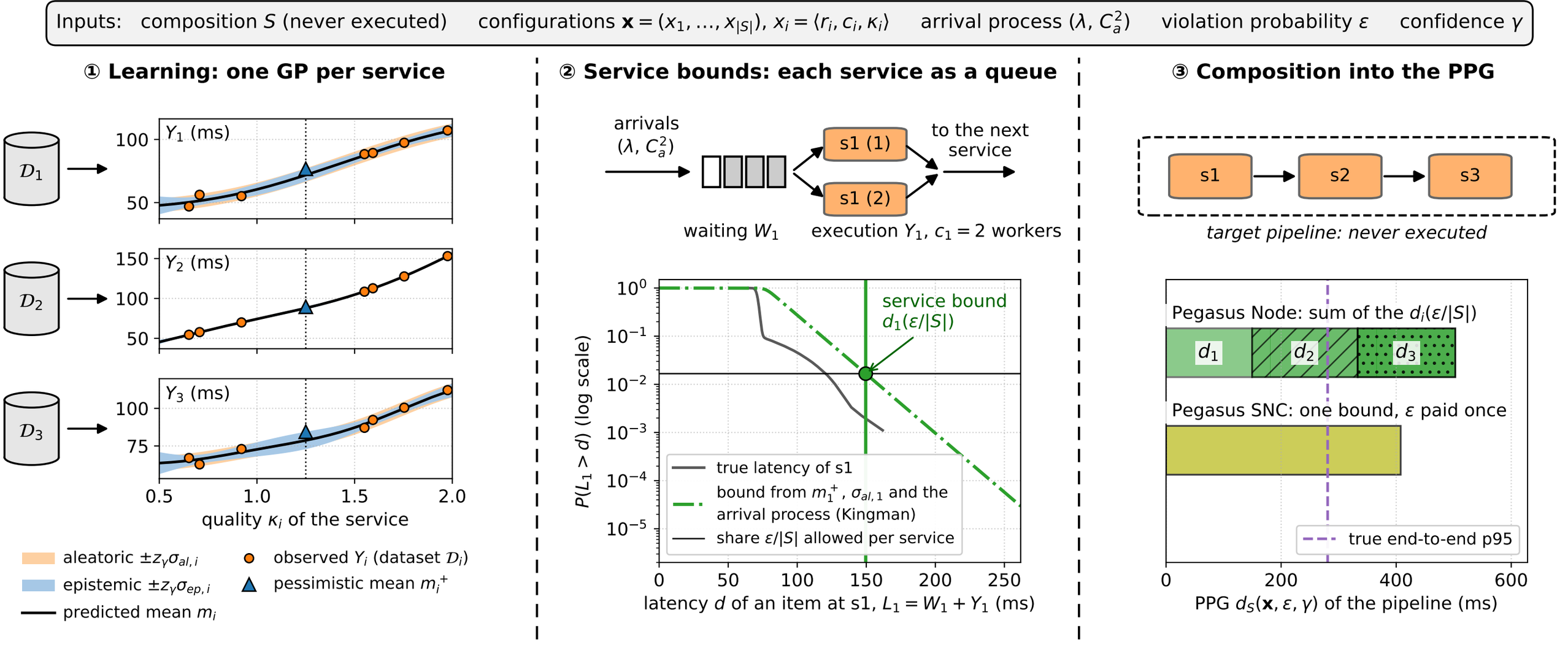

Inputs: composition S (never executed) configurations $\mathbf{x} = (x_1, \ldots, x_{|S|})$, $x_i = \langle r_i, c_i, \kappa_i \rangle$ arrival process (λ, C_a^2) violation probability ε confidence γ
① Learning: one GP per service
$\mathcal{D}_1$
$\mathcal{D}_2$
$\mathcal{D}_3$
Y_1 (ms)
Y_2 (ms)
Y_3 (ms)
100
50
150
100
50
100
75
0.5
1.0
1.5
2.0
quality κ_i of the service
aleatoric $\pm z_\gamma \sigma_{al,i}$
epistemic $\pm z_\gamma \sigma_{ep,i}$
predicted mean m_i
observed Y_i (dataset $\mathcal{D}_i$)
pessimistic mean m_i^+
② Service bounds: each service as a queue
arrivals
(λ, C_a^2)
s1 (1)
s1 (2)
to the next service
waiting W_1
execution Y_1, $c_1 = 2$ workers
service bound
$d_1(\varepsilon/|S|)$
$P(L_1 > d)$ (log scale)
10^0
10^{-1}
10^{-2}
10^{-3}
10^{-4}
10^{-5}
true latency of s1
bound from m_1^+, $\sigma_{al,1}$ and the arrival process (Kingman)
share $\varepsilon/|S|$ allowed per service
0
50
100
150
200
250
latency d of an item at s1, $L_1 = W_1 + Y_1$ (ms)
③ Composition into the PPG
s1
s2
s3
target pipeline: never executed
Pegasus Node: sum of the $d_i(\varepsilon/|S|)$
d_1
d_2
d_3
Pegasus SNC: one bound, ε paid once
true end-to-end p95
0
200
400
600
PPG $d_S(\mathbf{x}, \varepsilon, \gamma)$ of the pipeline (ms)